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MATHEMATICS

STANDARD SIX

First evaluation assessment - April 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	02E
Level	Standard Six
Year	2024
Exam Board / Issuer	VOPSO
Exam	First evaluation assessment - April
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts (8 main questions)

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S6-2024-VOPSO-April-Practice-assessment.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Section A: Mathematical operations

1(i). Calculate $99997 + 3$.

Step 1: 1 place: $7 + 3 = 10$; write 0, carry 1.

Step 2: 10 place: $9 + 0 + \text{carry } 1 = 10$; write 0, carry 1.

Step 3: 100 place: $9 + 0 + \text{carry } 1 = 10$; write 0, carry 1.

Step 4: 1000 place: $9 + 0 + \text{carry } 1 = 10$; write 0, carry 1.

Step 5: 10000 place: $9 + 0 + \text{carry } 1 = 10$; write 0, carry 1.

Step 6: Write final carry 1 in the next place.

Step 7: $99997 + 3 = 100000$.

Final Answer: 100000

1(ii). Calculate $4590 - 98$.

Step 1: 1 place: regroup 1 from the next place; $10 - 8 = 2$.

Step 2: 10 place: regroup 1 from the next place; $18 - 9 = 9$.

Step 3: 100 place: $4 - 0 = 4$.

Step 4: 1000 place: $4 - 0 = 4$.

Step 5: $4590 - 98 = 4492$; check $4492 + 98 = 4590$.

Final Answer: 4492

1(iii). Subtract 12 from -15.

Step 1: $-15 - 12$ means move 12 further units in the negative direction.

Step 2: $15 + 12 = 27$, so $-15 - 12 = -27$.

Final Answer: -27

1(iv). Calculate $-5 + 4$.

Step 1: The numbers have opposite signs; subtract their magnitudes: $5 - 4 = 1$.

Step 2: The greater magnitude belongs to -5 , so the result is -1 .

Final Answer: -1

1(v). Calculate $7004 \div 34$.

Step 1: $70 \div 34$: write 2; $2 \times 34 = 68$; $70 - 68 = 2$.

Step 2: $20 \div 34$: write 0; $0 \times 34 = 0$; $20 - 0 = 20$.

Step 3: $204 \div 34$: write 6; $6 \times 34 = 204$; $204 - 204 = 0$.

Step 4: $34 \times 206 = 7004$.

Step 5: $7004 \div 34 = 206$.

Final Answer: 206

1(vi). Multiply 3 km 125 m by 8.

Step 1: Convert to metres: $3 \times 1000 + 125 = 3125$ m.

Step 2: $3125 \times 8 = 25000$ m.

Step 3: $25000/1000 = 25$ km.

Final Answer: 25 km

1(vii). Simplify $2x + 3y - x + 6y$.

Step 1: Combine x terms: $2x - x = (2 - 1)x = x$.

Step 2: Combine y terms: $3y + 6y = (3 + 6)y = 9y$.

Step 3: Therefore the expression is $x + 9y$.

Final Answer: $x + 9y$

1(viii). Find the sum of the total values of digits 3 and 4 in 560.34.

Step 1: The value of 3 in the tenths place is $3/10 = 0.3$.

Step 2: The value of 4 in the hundredths place is $4/100 = 0.04$.

Step 3: $0.3 + 0.04 = 0.34$.

Final Answer: 0.34

1(ix). What is the place value of the underlined digit 0 in 50814?

Step 1: From right to left, the places are ones (4), tens (1), hundreds (8), thousands (0) and ten-thousands (5).

Step 2: The underlined 0 is in the thousands place.

Final Answer: Thousands

1(x). A school has 362 pupils. Write 362 in Roman numerals.

Step 1: $362 = 300 + 60 + 2$.

Step 2: $300 = \text{CCC}$, $60 = \text{LX}$ and $2 = \text{II}$.

Step 3: Combine them: CCCLXII.

Final Answer: CCCLXII

Section Summary: Align place values and decimal points. Convert mixed numbers to improper fractions; use a common denominator for addition and subtraction and the reciprocal for division.

Section B: Word problems

2(i). Asma did homework from 2140 hours to 2310 hours. How much time did she spend?

Step 1: Convert the times to minutes after midnight: $21 \times 60 + 40 = 1300$; $23 \times 60 + 10 = 1390$.

Step 2: $1390 - 1300 = 90$ minutes.

Step 3: 90 minutes = 1 hour 30 minutes.

Final Answer: 1 hour 30 minutes

2(ii). The head teacher distributed $\frac{8}{13}$ of the school library's 260 books. How many books were distributed?

Step 1: One thirteenth of 260 = $260/13 = 20$ books.

Step 2: Eight thirteenths = $20 \times 8 = 160$ books.

Final Answer: 160 books

2(iii). Write the number with 2 in millions, 7 in hundred-thousands, 1 in ten-thousands, 5 in hundreds and 6 in tens.

Step 1: The given place values are 2000000, 700000, 10000, 500 and 60.

Step 2: Use zero in the unmentioned thousands and ones places.

Step 3: $2000000 + 700000 + 10000 + 500 + 60 = 2710560$.

Final Answer: 2710560

2(iv). Bhoke multiplied a number by itself and obtained 841. Find the number.

Step 1: $29 \times 29 = 841$, so the expected positive whole number is 29.

Step 2: If negative integers are allowed, $(-29) \times (-29) = 841$ also.

Final Answer: 29 (also -29 if negative integers are allowed)

2(v). Find the GCF of the first two even numbers between 11 and 17.

Step 1: The first two even numbers are 12 and 14.

Step 2: Factors of 12 are 1, 2, 3, 4, 6, 12; factors of 14 are 1, 2, 7, 14.

Step 3: Their common factors are 1 and 2; the greatest is 2.

Final Answer: 2

2(vi). Asfia formed the smallest five-digit number with no repeated digits. Find that number.

Step 1: The first digit must be nonzero, so choose 1.

Step 2: For the remaining places choose the smallest unused digits in order: 0, 2, 3, 4.

Step 3: The result is 10234.

Final Answer: 10234

3(i). One cow has mass 150.008 kg. Find the mass in grams of 20 cows of the same size.

Step 1: $150.008 \times 20 = 3000.16$ kg.

Step 2: Since $1 \text{ kg} = 1000 \text{ g}$, $3000.16 \times 1000 = 3000160$ g.

Final Answer: 3000160 g

3(ii). The Mwanza-Shirati distance is 96.3 km and the Manila-Mbire distance is 102 km. Find their total distance.

Step 1: Align decimal places: $96.3 + 102.0$.

Step 2: $96.3 + 102 = 198.3$.

Final Answer: 198.3 km

3(iii). Mr. Majasho ran once around a rectangular garden measuring 300 m by 100 m. How far did he run?

Step 1: The distance is the perimeter: $2 \times (300 + 100)$.

Step 2: $300 + 100 = 400$; $2 \times 400 = 800$.

Final Answer: 800 m

4(i). Each matchbox contains 85 sticks. How many sticks are in 163 such boxes?

Step 1: Split into place values: $80 \times 163 = 13040$; $5 \times 163 = 815$.

Step 2: Add the partial products: 13855.

Final Answer: 13855 sticks

4(ii). Three school bells ring at intervals of 15, 20 and 30 minutes. After how many minutes will they next ring together?

Step 1: Find the LCM: $15 = 3 \times 5$; $20 = 2^2 \times 5$; $30 = 2 \times 3 \times 5$.

Step 2: $\text{LCM} = 2^2 \times 3 \times 5 = 60$.

Step 3: $60/15 = 4$, $60/20 = 3$ and $60/30 = 2$, so all three intervals divide 60.

Final Answer: 60 minutes

4(iii). Nindi's journey from the Earth to the Moon takes 30 days. Express this as weeks and days.

Step 1: One week has 7 days.

Step 2: $7 \times 4 = 28$; $30 - 28 = 2$.

Final Answer: 4 weeks 2 days

5(i). Convert 1.44 into a fraction.

Step 1: $1.44 = 144/100$.

Step 2: The numerator and denominator share factor 4: $(144/4)/(100/4) = 36/25$.

Step 3: As a mixed number, $36/25 = 1 \frac{11}{25}$.

Final Answer: $36/25$ ($1 \frac{11}{25}$)

5(ii). Round 17506410 to the nearest million.

Step 1: $17506410 = 17000000 + 506410$.

Step 2: 506410 is at least 500000, so round up from 17 million to 18 million.

Final Answer: 18000000

5(iii). A division has divisor 40 and dividend 8. Find the quotient.

Step 1: Quotient = dividend/divisor = $8/40$.

Step 2: Divide numerator and denominator by 8: $8/40 = 1/5 = 0.2$.

Final Answer: 0.2

6(i). Write $900 + 200000 + 3000 + 60 + 1000000$ in short form.

Step 1: Group the place values in descending order.

Step 2: $1000000 + 200000 + 3000 + 900 + 60 = 1203960$.

Final Answer: 1203960

6(ii). Sabrina has Sh 5000. She buys two pens at Sh 400 each and one book at Sh 1500. Find her remaining money.

Step 1: Pens cost $2 \times 400 = \text{Sh } 800$.

Step 2: Total spent = $800 + 1500 = \text{Sh } 2300$.

Step 3: Remaining = $5000 - 2300 = \text{Sh } 2700$.

Final Answer: Sh 2700

6(iii). A child is 120 months old. Find the age in years.

Step 1: There are 12 months in one year.

Step 2: $120/12 = 10$.

Final Answer: 10 years

Section Summary: Choose the operation from the question, convert to consistent units, and check the result. LCM uses the highest prime powers; GCF uses only factors shared by all numbers.

Section C: Geometry and statistics

7(i). The bar graph gives attendance for Monday, Tuesday, Wednesday, Thursday and Friday. Find the total attendance for the five days.

Step 1: Read the bars against the labelled scale: Monday 50, Tuesday 40, Wednesday 30, Thursday 50 and Friday 40.

Step 2: $50 + 40 + 30 + 50 + 40 = 210$.

Final Answer: 210 pupil attendances

7(ii). Five children weigh 6 kg, 2000 g, 7 kg, 5 kg and 12000 g. Find their average weight in grams.

Step 1: Convert kilograms to grams: 6 kg = 6000 g, 7 kg = 7000 g and 5 kg = 5000 g.

Step 2: Total = $6000 + 2000 + 7000 + 5000 + 12000 = 32000$ g.

Step 3: Average = $32000/5 = 6400$ g.

Final Answer: 6400 g

8(i). Triangle ABC is right-angled at C, with AB = 25 cm and CB = 24 cm. Find AC.

Step 1: By Pythagoras, $AC^2 = AB^2 - CB^2 = 25^2 - 24^2$.

Step 2: $625 - 576 = 49$.

Step 3: Since $7^2 = 49$ and a length is positive, $AC = 7$ cm.

Final Answer: 7 cm

8(ii). Name the type of angle shown with a small square at its vertex.

Step 1: The small square denotes perpendicular rays.

Step 2: The angle measures 90° , so it is a right angle.

Final Answer: Right angle

8(iii). A rectangle has perimeter 340 cm and length 96 cm. Find its area.

Step 1: $2 \times (\text{length} + \text{width}) = 340$, so $\text{length} + \text{width} = 340/2 = 170$ cm.

Step 2: $\text{Width} = 170 - 96 = 74$ cm.

Step 3: $\text{Area} = 96 \times 74 = 7104 \text{ cm}^2$.

Final Answer: 7104 cm²

Section Summary: Use marked measurements and angles. Circle sectors are fractions of 360° . Right triangles obey $a^2 + b^2 = c^2$; cylinder volume is $\pi \times r^2 \times h$. Read graph coordinates as (x, y).

Source Checks

Darasa Huru - VOPSO Standard Six Assessment Examination April 2024